

Considered there are N philosophers seated around a circular table with one chopstick between each pair of philosophers. There is one chopstick between each philosopher. A philosopher may eat if he can pick up the two chopsticks adjacent to him. One chopstick may be picked up by any one of its adjacent followers but not both. Write a program to solve the problem using process synchronization technique.

```
/c/Users/acom/Desktop/OS_CD24037
GNU nano 8.7
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5

sem_t chopstick[N];

void* philosopher(void* num) {
    int id = *(int*)num;

    printf("Philosopher %d is thinking\n", id);
    sleep(1);

    // Deadlock prevention
    if (id == N - 1) {
        sem_wait(&chopstick[(id + 1) % N]);
        sem_wait(&chopstick[id]);
    } else {
        sem_wait(&chopstick[id]);
        sem_wait(&chopstick[(id + 1) % N]);
    }

    printf("Philosopher %d is eating\n", id);
    sleep(2);

    sem_post(&chopstick[id]);
    sem_post(&chopstick[(id + 1) % N]);

    printf("Philosopher %d finished eating\n", id);
    return NULL;
}

int main() {
    pthread_t phil[N];
    int ids[N];

    for (int i = 0; i < N; i++) {
        sem_init(&chopstick[i], 0, 1);
    }

    for (int i = 0; i < N; i++) {
        ids[i] = i;
        pthread_create(&phil[i], NULL, philosopher, &ids[i]);
    }

    for (int i = 0; i < N; i++) {
        pthread_join(phil[i], NULL);
    }

    return 0;
}
```

 /c/Users/acom/Desktop/OS_CD24037

```
acom@DESKTOP-50T9G9Q MSYS ~  
$ cd /c/Users/acom/Desktop/OS_CD24037  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$ nano phi.c  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$ gcc phi.c -o phi  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$ ./phi  
Philosopher 0 is thinking  
Philosopher 1 is thinking  
Philosopher 2 is thinking  
Philosopher 3 is thinking  
Philosopher 4 is thinking  
Philosopher 4 is eating  
Philosopher 3 is eating  
Philosopher 4 finished eating  
Philosopher 2 is eating  
Philosopher 3 finished eating  
Philosopher 1 is eating  
Philosopher 2 finished eating  
Philosopher 1 finished eating  
Philosopher 0 is eating  
Philosopher 0 finished eating  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$
```